Lecture 1.7: Functions in JavaScript

What is a function?

A **function** is a reusable block of code that performs a task. Functions help you:

- avoid repeating code
- organize logic into smaller pieces
- make code easier to test and maintain

Basic function syntax

```
function add(a, b) {
    return a + b;
}
```

Validation and defensive coding (from your lecture)

This function checks if parameters were provided, converts them to numbers, and validates them:

```
function add(a, b) {
    if (a === undefined || b === undefined) {
        console.error("Values not provided");
        return;
    }

    a = parseFloat(a);
    b = parseFloat(b);

    if (isNaN(a) || isNaN(b)) {
        console.error("Invalid number");
    } else {
```


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```
        console.log(a + b);
    }
}
```

> **Instructor insight:** A function is most useful when it **returns** a value instead of printing it. Printing is for debugging; returning is for building real programs.

Default parameters (ES6)

Default values are used so missing inputs do not cause `undefined` issues:

```
function add2(a = 0, b = 0) {
    a = parseFloat(a);
    b = parseFloat(b);

    if (isNaN(a) || isNaN(b)) {
        console.error("Invalid number");
    } else {
        console.log(a + b);
    }
}
```

Returning values (production style)

This is the refined version of `add3` idea: return a number (or `NaN` when invalid), and let the caller decide what to do.

```
function add3(a = 0, b = 0) {
    a = parseFloat(a);
    b = parseFloat(b);

    if (isNaN(a) || isNaN(b)) {
        return NaN;
    }

    return a + b;
}

var result = add3(45, 25);
```


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```
if (isNaN(result)) {
    console.error("Invalid input");
} else {
    console.log(result);
}
```

Output:

`70`

Area of a circle

task: accept a radius and return area. Formula: $area = \pi r^2$.

```
function circleArea(radius) {
    var r = parseFloat(radius);

    if (isNaN(r) || r < 0) {
        return NaN;
    }

    return Math.PI * r * r;
}

console.log(circleArea(5));
console.log(circleArea("abc"));
```

Output:

For `5` → `78.53981633974483`

For `"abc"` → `NaN`

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Common mistakes with functions

- Forgetting `return` when you need a value.
- Doing input/output inside the function (makes it hard to reuse).
- Not validating user input.

Practice tasks

1. Create a function `max(a, b)` that returns the bigger number (handle invalid inputs).
2. Create a function `isEven(n)` that returns `true` or `false`.
3. Create a function `discountedPrice(amount, isVip)` and return the final amount.

JS Refined Notes — Lecture 1.7

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